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# Can Entrepreneurial Initiative Blunt the Economic Inequality–Growth Curse? Evidence From 92 Countries

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## Abstract

Despite the growing interest in understanding the effects of income inequality on economic growth, the influence of entrepreneurship-related institutional constraints on the inequality–growth association remains less understood. Drawing on an institutional constraints perspective in the context of startup entry regulation and credit constraints, we propose that under increasing income inequality, ease of startup or access to credit from the financial sector is positively associated with per capita economic growth. In a sample of 92 countries, robust to alternate specifications, we find support for our hypotheses. Our findings contribute to the debates about the income inequality–growth nexus and imply potential policy interventions in the form of lower startup regulations and domestic credit provision from the financial sector to enhance economic growth under increasing economic inequality.

## Keywords

entrepreneurship, inequality, new business formation

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Income inequality is a persistent and vexing challenge that has existed for thousands of years. Indeed, "modern" inequality can be traced to the evolution of human survival from foraging to farming more than 10,000 years ago (Atkinson & Bourguignon, 2014; Brenner, Kaelble, & Thomas, 1991). Since that time, nearly every society has wrestled with the social and economic costs and benefits of inequality. Most recently, Thomas Piketty's (2014) book, *Capital in the 21st Century*, has renewed concerns about the impact of various economic policies on growing inequality and has underscored the divisive and corrosive aspects of inequality in modern society. Theoretical discourses on lowering the negative effects of income inequality point to the importance of (a) improving circumstances of the poor, (b) lowering income gaps between individuals, (c) reducing wealth concentration, and (d) increasing the size of the middle class and improving its circumstances (Klasen, 2016). Beyond the obvious concerns about fairness, equity, and justice, several studies have focused on the question of whether economic inequality is associated with economic growth[1] (see Benabou, 1997, for a review).

To better understand the inequality–growth relationship, we turn to the literature on institutional constraints (Child & Tsai, 2005; Clingermayer & Feiock, 2014; Witt & Lewin, 2007). We suggest that institutional constraints could contribute to inefficiency in the allocation of societal and economic resources driven by inequality and could further reinforce systems, roles, structures, and social and economic discourses spawned by income inequality. Lowering constraints that impede new firm entry may, therefore, enhance growth under increasing inequality. The institutional constraints framework is rooted in Baumol's (1990) work, in which he proposes that institutional constraints may limit productive, and foster unproductive, entrepreneurial activities. Sobel (2008) tested Baumol's framework and found that "good institutions channel effort into productive entrepreneurship, sustaining higher rates of economic growth" (p. 641). Lowering institutional constraints to entrepreneurship could unleash the forces of creative destruction, promote organic growth, lower wealth concentration, and lower disparity in access to, and exploitation of, social and economic opportunities (Wennekers & Thurik, 1999).

Institutional constraints to entrepreneurial activities have been especially salient in recent years with stagnating wages and increasing wealth concentration (Dabla-Norris, Kochhar, Suphaphiphat, Ricka, & Tsounta, 2015). Relaxing institutional constraints to entrepreneurial activity may weaken rigidities that undergird inequality by improving access to opportunities; challenging the structures, norms, and discourses that sustain inequality; removing incentives for rent seeking from incumbents and public officials; and weakening the institutions that sustain the pervasive effects of inequality.

These outcomes could, in turn, increase growth even in an environment of increasing income inequality. Acs, Desai, and Hessels (2008) have also called for more research at the intersection of entrepreneurship, economic development, and institutional quality. An emerging body of research in entrepreneurship has highlighted the constraining role of inequality in the entrepreneurial process. Inequality lowers marginal returns from human and financial capital in entrepreneurial efforts (Xavier-Oliveira, Laplume, & Pathak, 2015) and increases the threshold to achieve self-employment (Sarkar, Rufín, & Haughton, 2018). More broadly, inequality influences firms and markets by stimulating "(1) social movements that coerce and constrain firms' actions, (2) alternative organizational forms that displace existing organizations, and (3) new political and regulatory risks that undermine firms' performance or survival" (Bapuji & Neville, 2015, p. 233). Furthermore, the decline of traditional industrial corporations and related trends of outsourcing and erosion of union participation have eroded employee safety net, driving higher income inequality and lowering upward mobility (Davis, 2016). Taking a configurational perspective, Lewellyn (2018) concludes that "high-growth and/or necessity [-based entrepreneurial activity] when occurring in a given context with particular institutional complementarities is not sufficient to prevent income inequality; rather in some cases, it is helping to drive the widening income gap between different societal actors" (p. 1143). More recent studies show that removing institutional constraints could help overcome the constraining effects of inequality. Culturally endorsed transformational leadership activities could help overcome constraining effects of inequality to promote social entrepreneurship (Muralidharan & Pathak, 2018), and immigrant entrepreneurs can overcome constraints from inequality when they challenge socially constructed boundaries (Griffin-EL & Olabisi, 2018). Building on this emerging body of research at the intersection of inequality and entrepreneurship, we expect that under increasing income inequality, reducing entrepreneurship-related institutional constraints could stimulate income growth.

Building on two central theories related to institutional constraints—entry regulation theory (de Soto, 1989) and credit constraints framework (Black & Strahan, 2002)—we extend normative research that has sought to inform how inequality can be reduced but has not explored how the often-inevitable presence of inequality and its impacts can be partially offset by factors promoting vibrant entrepreneurial activity. Specifically, we ask the following research question: With increasing income inequality, can lower institutional constraints to entrepreneurial activities, such as ease of startup[2] or access to credit from the financial sector, drive growth per capita? (Arum & Müller, 2009).

Drawing on a sample of 92 countries from 2004 to 2014, we find that with increasing income inequality, ease of startup or access to credit are positively associated with growth per capita. The findings are robust to alternative specifications and a test that addresses endogeneity concerns.

The rest of the article is organized as follows. In the next section, we review the literature on inequality and growth. We then develop our theoretical perspective on the role of institutional constraints—ease of startup and credit from the financial sector. We then describe our empirical setting, data, and modeling. Next, we report our results and examine them considering broader questions about inequality, growth, and income. We conclude with implications of our research, limitations, and suggestions for potentially interesting extensions of our findings.

## Inequality, Entrepreneurship, and Growth: Theory and Hypotheses

Over the past three decades, the interplay between income inequality and growth has attracted considerable attention from economists and social scientists (Atkinson & Bourguignon, 2014; Brenner et al., 1991). According to Bapuji and Mishra (2015), economic inequality encompasses disparity in income receipts (income inequality) and disparity in the distribution of financial wealth (wealth inequality). Distinguishing economic inequality from poverty, Bapuji and Mishra (2015) state that "poverty is about meeting basic needs, whereas economic inequality places in focus the unevenness in the distribution of income and wealth, irrespective of the presence and absence of poverty" (p. 441). National levels of inequality also have multilevel and multicontextual influences within a country (Edwards, 2015; Greenwood, Oliver, Lawrence, & Meyer, 2017; Suddaby, Bruton, & Walsh, 2018). Studies on inequality in organizational research have evolved over four waves (Davis, 2017), starting with the first wave on stratification research in sociology during the 1960s to 1970s and the second wave aimed toward understanding recruitment and promotion-related inequality. The third wave focused on equal opportunity practices and the most recent wave has focused on how institutions influence employment policies and outcomes (Davis, 2017). Overall, the relationship between economic growth and income inequality has been an important research agenda in economics and has recently gained momentum in organizational research. Next, we provide a brief overview of the relationship between income inequality and economic growth.

## *Inequality and Growth*

Perhaps, the most well-known scholar who hypothesized the relationship between inequality and growth was Simon Kuznets, who, in the mid-1950s, suggested that in the early stages of development, income inequality would increase, plateau in the middle stages, and decline once countries reached higher income status (Kuznets, 1955). The explanation for this process was complex but generally hinged on the progression of economies from agricultural to industrial, and then postindustrial, in which democratization and rise of the welfare state would allow the benefits of growth to be distributed among broader income strata as per capita income grew. Using a somewhat different logic, Kaldor (1955) came to essentially the same conclusion.

In a 1997 review, Benabou concluded that the relationship between initial inequality and growth is negative but that the relationship in the face of higher levels of development is less clear. Despite an emerging consensus in the mid- and late 1990s, there have been some countervailing views, notably Forbes (2000), who finds that in the short and medium term, an increase in a country's level of income inequality has a *positive* relationship with subsequent economic growth. More recently, Ezcurra (2007) examined the relationship between income inequality and economic growth using multiple measures and extensive controls in the regions of several European Union countries over the period 1993 to 2002, and finds that income dispersion was negatively associated with regional growth. Building on prior empirical work on inequality and growth, Herzer and Vollmer (2013) employed a panel of nine high-income countries over the period from 1961 to 1996 and found that inequality lowers growth, and in the long run, economic growth increases inequality.

Over the years, several studies have used a variety of theoretical and empirical approaches to explore this relationship between inequality and growth. We attempt to present a partial summary of this large literature in Table 1.

Reviewing empirical work on the inequality–growth relationship, Neves and Silva (2014) concluded that the mixed results were due to "differences in the type of countries and time periods included in the samples, the variable used to measure inequality, the structure of the data, and the estimation techniques" (p. 1) and that the inequality–growth association is highly contextual. Klasen (2016) infers that studies finding a positive association are generally cross sectional, but "more and more papers come to the conclusion that lower inequality is more conducive to growth" (p. 65). He concluded that the relationship can be positive, negative, decreasing, or not at all significant, depending on the specification (Klasen, 2016). According to Halter, Oechslin,

**Table 1.** Survey of Literature on the Relationship Between Inequality and Growth.

| Study | Independent variable | Dependent variable | Finding |
|---|---|---|---|
| Ezcurra (2007) | Personal income inequality | Regional economic growth in the European Union | The degree of income dispersion is negatively associated with regional growth. |
| Bourguignon (2017) | | | The inequality of income, on which the empirical literature focuses almost exclusively, is an imperfect marker of their broader definition of inequality. |
| Sbardella, Pugliese, and Pietronero (2017) | Development and industrialization | Wage inequality | Countries with an average level of development suffer the highest levels of wage inequality. |
| Thomas (2015) | Income inequality (distribution) | Economic growth | Finds potential correlation between income distribution and economic growth rates. |
| Sehic and Huskic (2015) | Inequality | Economic development | A statistically significant negative relationship between higher initial inequality and economic development during the period 1990-2010. |
| Akaev, Ichkitidze, Sarygulov, and Sokolov (2016) | Economic inequality | Regional development | Increasing polarization in the income levels of the various regions acts as the dominating trend of growing economic inequality. |
| Prete (2013) | Financial development | Income inequality moderated by economic literacy | The empirical association between financial development and lower income inequality appears to be driven by economic literacy. |
| Nigar (2015) | Inequality | Economic growth | Inequality can adversely affect the otherwise positive impact of institutional quality on growth. |
| Iniguez-Montiel (2014) | Income inequality | Economic growth | For Mexico, a middle-income country exhibiting quite low growth rates and high inequality levels, a further improvement in its distribution of income and assets is essential if the economy is to succeed in making a real dent in poverty. |

*(continued)*

**Table 1. (continued)**

| Study | Independent variable | Dependent variable | Finding |
|---|---|---|---|
| Roe and Siegel (2011) | Inequality | Economic development | Inequality-perpetuating conditions that result in political instability and weak democracy are fundamental roadblocks for international organizations such as the World Bank that seek to promote financial development. |
| Fielding and Torres (2006) | Inequality | Economic development | The causal relationship runs from income inequality to income and life expectancy. |
| Valerio Mendoza (2014) | Economic and technological development | Income inequality | The spatial decomposition of inequality shows that cities with a zone, whether it is an ETDZs or HIDZs have higher incomes and lower income inequality than cities without any zone. |
| Savvides and Stengos (2000) | Income inequality | Economic development | Using a threshold regression model that allows for endogenous income thresholds separating developed from less developed countries, income inequality and per capita income association depend on the level of economic development. |
| Shuai (2015) | Income inequality | Economic growth and county-level job attractiveness | State and local governments' efforts to attract and retain manufacturing jobs help improve county-level income distribution, whereas attracting jobs in professional and business services worsen county-level income inequality. |
| Nel (2015) | Inequality | Economic development | Regional and local demographic and economic inequality increased due to demographic decline and higher contraction in relative economic share compared with cities and selected regions. |

*(continued)*

**Table 1. (continued)**

| Study | Independent variable | Dependent variable | Finding |
|---|---|---|---|
| Delbianco, Dabús, and Caraballo (2014) | Income inequality (distribution) | Economic growth | Progressive redistributive policies in favor of poorer layers of population promote economic growth in lower income economies. |
| Iyigun and Owen (2004) | Income inequality | Economic development | The level of financial development may help explain why the distribution of income affects the short-run variability of consumption and output differently in high-income countries than in low-income countries. |
| Sylla (2015) | Income inequality | Economic growth | Increasing nations' and households' access to finance can reduce inequalities, but that will require more financial education and financial literacy, which at present appear to be woefully inadequate. |
| Bittencourt (2010) | Income inequality | Economic development | A deeper and more active financial sector alleviates the high inequality seen in Brazil without the need for distortionary taxation. |
| Hunter (2001) | Economic inequality | Economic development | Inequality is not associated with intergenerational wealth mobility across classes in stagnant economies. |
| Akinci (2017) | Inequality | Economic growth | An increase in the incomes of the rich raises the incomes of the poor and vice versa. However, the contribution of the income transfer of the poor to the rich is more dominant. |
| Ram (1997) | Income inequality | Economic development | Inequality does not decline with an increase in income even at high levels of development. |
| Easterly (2007) | Income inequality | Development | Agricultural endowments predict inequality and inequality predicts development. |

*(continued)*

**Table 1. (continued)**

| Study | Independent variable | Dependent variable | Finding |
|---|---|---|---|
| Galor and Moav (2004) | Inequality | Development | As human capital emerged as a growth engine, equality alleviated the adverse effects of credit constraints on human capital accumulation, stimulating the growth process. |
| Braun (1991) | Income inequality | Economic development | Income inequality results in widely divergent effects among regions. |
| Jauch and Watzka (2016) | Income inequality | Financial development | Financial development increases income inequality. |
| Trejos (2000) | Income inequality | Economic growth | Income inequality is a precondition to contemporaneous economic growth in both the medium and the short run. |
| Howarth and Kennedy (2016) | Inequality | Economic growth | A growth rate of 1.2% per year arises when income is adjusted to account for the social costs of inequality even though the growth in after-tax income per capita for the United States between 1979 and 2011 was 1.7%. |
| Lorenzi (2016) | Inequality | Economic growth | Political freedom, property rights, the rule of law, education, social mobility, and wealth transfers can facilitate economic growth and equality. |
| Majeed (2016) | Inequality | Economic growth and trade | High initial inequalities in developing countries are the likely reason for a negative relationship between trade and economic growth. |
| Ncube, Anyanwu, and Hausken (2014) | Inequality | Economic growth and poverty | Income inequality reduces economic growth and increases poverty in the region. |
| Neagu, Dumiter, and Braica (2016) | Income inequality | Growth | GDP per capita and inequality are positively associated, the inward stock of FDI and market capitalization further increase inequality, and education levels lower inequality. |

*(continued)*

**Table 1. (continued)**

| Study | Independent variable | Dependent variable | Finding |
|---|---|---|---|
| Kefi and Zouhaier (2012) | Inequality | Economic growth | There is a negative effect of inequality on economic growth and vice versa. |
| Adams (2004) | Inequality and poverty | Economic growth | Although economic growth does reduce poverty in developing countries, the rate of poverty reduction depends very much on how economic growth is defined. |
| Hasanov and Izraeli (2011) | Income inequality | Economic growth | Lowering inequality or increasing it substantially reduces growth; thus, stable inequality may be good for growth. |
| Turnovsky (2015) | Inequality | Economic growth | The growth inequality association is contingent on the analytical approach used and within a given approach the assumptions of the relative magnitude of externalities, underlying financing policies, the time period under consideration, and factor substitutability can further influence this association. |
| Andrei (2015) | Inequality | Economic growth | In a review of different models, a broad conclusion is that inequality reduces growth. |
| Schober and Winter-Ebmer (2011) | Gender wage gaps | Economic growth | Gender inequality is negative for growth. |
| Risso and Sánchez Carrera (2012) | Inequality | Economic growth | In China, the long-run causal relationship between economic growth and income inequality in China during the prereform (1952-1978) was unidirectional from inequality to growth; however, there was no directional causality in the postreform (1979-2007) period. |
| Aghion, Caroli, and Garcia-Penalosa (1999) | Inequality | Economic growth | A one-time reduction in after-tax inequality that would foster investment incentives and growth in the short run would result in a (possibly temporary) upsurge in inequality because of the accelerated technical progress it induces. |

*(continued)*

**Table 1. (continued)**

| Study | Independent variable | Dependent variable | Finding |
|---|---|---|---|
| Fallah and Partridge (2007) | Inequality | Economic growth | When separately considering metropolitan and nonmetropolitan samples, there is a positive inequality–growth link in the urban sample. |
| Shahbaz (2010) | Income inequality | Economic growth | Support for the existence of Kuznets inverted-U as well as an inverted S-shaped curve in Pakistan. |
| Vu and Mukhopadhaya (2011) | Inequality | Economic growth | The impact of inequality on economic growth is negative for a sample of 74 developed and developing countries, with larger negative effects in low-income countries. |
| Iceland (2003) | Economic inequality | Economic growth and poverty | Income growth explains most of the trends in absolute poverty, whereas inequality generally plays the most significant role in explaining trends in relative poverty. |
| Pinney (2014) | Inequality | Economic growth | Avoiding going back to a pre–20th-century norm of high-income inequality and low economic growth may require a fundamental rethinking of the role of employment as the primary mechanism for income distribution, especially in a world in which the concept of work will be radically redefined. |
| Tello and Ramos (2012) | Income inequality | Economic growth | Spatial idiosyncrasies and context limit "one size fits all" policy recommendations for managing regional inequality and growth in Mexican regions. |
| Hui (2013) | Inequality | Economic growth | Minimum wage along with in-work benefit schemes, social safety net, could lower inequality of low-income workers in Singapore. |
| McDonald and Zhang (2012) | Income inequality | Economic growth | Initial inequality has a positive indirect effect on average output growth by lowering the ratio of physical to human capital, besides its standard negative direct effect. |

*(continued)*

**Table 1. (continued)**

| Study | Independent variable | Dependent variable | Finding |
|---|---|---|---|
| Coatsworth (2008) | Inequality | Economic growth in Latin America | The contrasting view offered here affirms the significance of institutional constraints but argues that they did not arise from colonial inequalities, but from the adaptation of Iberian practices to the American colonies under conditions of imperial weakness. |
| Fanta and Upadhyay (2009) | Income inequality | Economic growth | Poverty decreases in response to economic growth, with the estimated elasticity ranging between −0.5 and −1.10. |

*Note.* ETDZ = economic and technological development zones; HIDZ = high-tech industrial development zones; FDI = foreign direct investment.

and Zweimüller (2014) and Ostry, Berg, and Tsangarides (2014), estimates based on fixed effects or first differences (Forbes, 2000) pickup effects that are positive in the short run, whereas studies drawing on *both* the cross-sectional and time-series variation find support for a negative association between inequality and growth.

Overall, the mixed findings on the inequality–growth association could be driven by a variety of factors, including institutional contingencies, sampling heterogeneity, differences in the operationalization of measures, and empirical specifications. Addressing these empirical complexities is beyond the scope of this article; nevertheless, studies aimed at reconciling these empirical complexities and accounting for institutional contingencies are critical to resolving the debate on the directionality of the inequality–growth association and providing policy prescriptions. We instead focus on how alleviating institutional constraints on new firm entry and improving credit access from the financial sector could drive economic growth under increasing income inequality.

## *Institutional Constraints—Entry Regulations and Credit Access*

Defining institutions as a mode of control, Commons (1931) was one of the first to highlight the constraining effect of institutions. Ingram and Clay (2000), from a neoinstitutional perspective, define institutions as a set of rules that constrain individuals and organizations from conducting their activities. As explained by Khoury and Prasad (2016), institutional constraints can lead to disadvantageous socioeconomic arrangement and could restrict economically beneficial behaviors:

> Country environments with formal institutional constraints can be represented by numerous conditions including ineffectual rule of law, government corruption, exploitation of public funds/resources, the selective application or enforcement of law, inequitable systems of justice according to group-affiliation, limited access to education or public benefits, constrained civil liberties, restricted international trade or market-entry, presence of state-controlled media, or the ongoing risk of political hazards. (p. 937)

As collective and regulatory political and social structures, institutions impose laws, rules, and norms that levy constraints on entrepreneurial activities (North, 1990). Based on Scott (1995), entrepreneurship-related institutional constraints emerge from the three institutional pillars—regulative pillar (or, the formal rules and enforcement mechanisms for new firm entry), normative pillar (or, the socially acceptable means through which individuals should pursue entrepreneurial activities), and cognitive pillar (or, internalized beliefs and values about entrepreneurship).

We focus on the regulative pillar that imposes institutional constraints on entrepreneurial activity. Regulative institutional constraints are central to shaping entrepreneurial activities by formulating and imposing startup policies and indirectly influencing cultural norms and behavioral and social expectations toward entrepreneurial activities (Aldrich, 1990). Regulative constraints to entrepreneurship include legal impediments to starting a business, restrictive fiscal policies that limit access to capital and increase cost of doing business, among others (Bruton, Ketchen, & Ireland, 2013; de Soto, 1989; Nyström, 2008). Indirectly highlighting the potential role of entrepreneurship-related institutional constraints in driving economic growth, differences in entrepreneurship rates explain one third (Reynolds, 1999) to a half (Zacharakis, Bygrave, & Shepherd, 2000) of differences in national economic growth rates. Institutional constraints also explain the degree of entrepreneurial transformation in former Soviet countries in central Asia (Luthans, Stajkovic, & Ibrayeva, 2000). The regulative constraints we focus on in our work influence drive and motivation of entrepreneurs. For example, Branstetter, Lima, Taylor, and Venâncio (2014) found that relaxing firm registration procedures led to higher new firm entry in Portugal. Similarly, reviews have found that greater credit availability also promotes entrepreneurial growth (Black & Strahan, 2002; Demirgüç-Kunt & Singer, 2017; Fraser, Bhaumik, & Wright, 2015).

Using the above literature as the backdrop, the two widely studied institutional constraints to new firm entry and exit are entry regulation theory (de Soto, 1989) and credit constraints framework (Beck, Büyükkarabacak, Rioja, & Valev, 2012; Hutchinson & Xavier, 2006; Schumpeter, 1912/1934). In transitioning from the nascent stage to the startup stage, ease of startup procedures can be critical. For both startups and small established firms, access to credit from the financial sector could improve survival and the growth odds of new and/or small firms. These two institutional constraints undergird the vibrancy of institutions in promoting, facilitating, and sustaining entrepreneurial activity.

*Ease of startup, income inequality, and economic growth.* Ease of startup—the number of procedures, official time to complete the procedures, and official costs for completing the procedures—refers to the stringency of firm entry regulations. According to the World Bank Indicator data,[3] the average cost of business startup procedures was the lowest in North America (0.8%) and the highest in Africa (25.2%). Worldwide, the average cost of business startup procedures declined from 11.2% in 2003 to 3.6% in 2016 (Doing Business Report, 2017). South Asia and East Asia and Pacific registered the highest decline in business startup costs between 2003 and 2016 from 47.3% to 13.4% and from 51.2% to 17.2%, respectively (Doing Business Report, 2017). Although the cost of business startup has declined over the past decade, substantial business startup costs could dissuade entrepreneurial activity.

Benefits and costs of entry regulation can be further understood through the lens of public interest theory and public choice theory (Djankov, La Porta, Lopez-de-Silanes, & Shleifer, 2002). According to Pigou's (1938) public interest theory, baseline regulations are necessary to reduce market failure. Some regulations are necessary because zero startup procedures would increase negative externalities by allowing entry of undesirable firms who are more likely to fail. Unchecked entry of less desirable firms could lower the quality of products or services, increase economic costs from business failures, and lower overall societal welfare. Conversely, public choice theory by Tullock (1967) states that government regulations are inefficient and entry regulations are "designed and operated primarily for" benefits to the incumbents (Stigler, 1971, p. 3), reducing competition and increasing incumbent profits (Kaplan, Piedra, & Seira, 2011). Jointly considering both, public interest and public choice theory provide a more viable middle ground, where ease of startup ensures "that new companies meet minimum standards to provide a good or service. By being registered, new companies acquire a type of official approval, which makes them reputable enough to engage in transactions with the general public and other businesses" (SRI International, 1999, p. 14). Despite the potential benefits and costs of higher entry regulation, the empirical literature points to its negative effects.

de Soto's (1989) work on entry regulation theory found that bureaucratic costs and corruption were the key hindrances in the launch of formal businesses in Peru. Lower ease in startup procedures is rooted in the tollbooth theory, where regulating new firm entry benefits bureaucrats and politicians through rent seeking and rent generation (Djankov et al., 2002). According to Shleifer and Vishny (1993), "An important reason why many of these permits and regulations exist is probably to give officials the power to deny them and to collect bribes in return for providing the permits" (p. 601). Politicians and regulators may actively increase startup regulations to sustain cronyism, secure votes, or receive political contributions from interest groups. Higher entry regulation for new businesses invariably leads to worse outcomes for potential entrepreneurs (Dreher & Gassebner, 2013; Levie & Autio, 2011).

Related to the empirical support for the entry regulation theory, based on a 10-year panel data from 180 countries, Divanbeigi and Ramalho (2015) find that an improvement of "10 points in the overall measure of business regulations is linked to an increase of around 0.5 new businesses per 1,000 adults" (p. 2). Klapper and Love (2011) found support for a positive relationship between lower business startup regulation and firm creation in a cross-country sample. Analyzing the 2005 "On the Spot" program in Portugal—a program that significantly reduced startup costs—Branstetter and colleagues (2014) found that during the first 2 years 4,500 new, mostly small, firms were launched and 17,500 new jobs were created, whereas in a quasi-natural experimental setup, Bruhn (2011) found that a simplified registration process in Mexico increased both firm creation and growth.

Under increasing inequality, ease of startup could be positively related to growth for the following reasons. First, access to economic opportunities could restrain the clutch of inequality on growth. New firm entry, as a source of creative destruction, increases jobs, promotes competition and innovation, and contributes to economic growth (Aghion, Blundell, Griffith, Howitt, & Prantl, 2009; Ciccone & Papaioannou, 2007; Haltiwanger, Jarmin, & Miranda, 2013; Schumpeter, 1912/1934). The vibrancy of new and small firms has been linked to high levels of job creation (McDermott & Mejstrik, 1992) and long-term social welfare through macroeconomic development (FitzRoy, 1993). Under increasing inequality, lower startup ease would restrict individual mobility and economic security (Mair, Martí, & Ventresca, 2012; Moser & McIlwaine, 2006), further increasing the vise of income inequality on growth.

New firm entry has a twofold effect: first, entrepreneurs with residual rights to the profits are able to increase their own income; and second, the regional and industry spillovers that result from entrepreneurial activity increase overall growth (Ghio, Guerini, Lehmann, & Rossi-Lamastra, 2015; Katz & Krueger, 2017). Higher entry regulation could strengthen the grip of inequality by lowering job growth and competition (Ciccone & Papaioannou, 2007) and exacerbate industry concentration (Fisman & Allende, 2010; Klapper, Laeven, & Rajan, 2006). Divanbeigi and Ramalho (2015) found that "moving from the lowest quartile of improvement in business regulations to the highest quartile is associated with a significant increase in annual per capita growth of around 0.8 percentage points" (pp. 4-5). Along these lines, as the degree of entry regulation declines, new ease of startup would reduce the effects of inequality that restrain pursuit of entrepreneurial opportunities.

Second, industry incumbents who may be less efficient and/or competitive under inequality, with a higher ease of startup may become more competitive and reduce rent seeking. As ease of startup challenges the existing modes of doing business, the insulating mechanisms of inequality for the incumbents may become less resistant to competition, requiring incumbents to invest in growth opportunities. In extreme cases, firms such as Airbnb and Uber may fundamentally challenge industry structures to facilitate growth through entrepreneurship. Therefore, new firm entry could improve markets both directly (through new business models and improving access to opportunities) and indirectly (by increasing competition and resource allocation efficiency).

Third, ease of entry further weakens the grip of income inequality on growth by providing the sociocultural infrastructure and improving the overall entrepreneurial climate in a country. Lower institutional frictions in starting a business signal growing support of the government in promoting competition and improving growth through grassroots entrepreneurship. Ease of startup could empower more marginalized groups to enter commerce and pursue self-realization goals. Ease of startup also lowers corruption and improves transparency to lower rent seeking from officials, thus channeling resources toward value-creating activities. As a force of creative destruction, a higher rate of entry is indicative of the efficacy of property rights infrastructure that improves transparency in allocation and exploitation of resources, and primes pursuit of growth opportunities from all strata.

Overall, we expect that as income inequality increases, lower institutional constraints to promote ease of startup will be positively associated with growth.

**Hypothesis 1:** With increasing income inequality, greater ease of startup will be positively associated with GDP per capita growth.

*Credit availability, provision, income inequality, and economic growth.* Based on the credit constraints framework (Black & Strahan, 2002), availability of credit from the financial sector promotes vibrant credit markets necessary to exploit economic opportunities. Availability of credit from the private sector, representative of the infrastructure for financial inclusion of small firms, lowers bankruptcy and improves growth (Beck & Levine, 2003). Supporting the role of access to credit for the firm establishment and firm continuity stages, Lindh and Ohlsson (1998) note that "liquidity constraints are binding on the decision to become and stay self-employed" (p. 45). Both startups and small and medium-sized firms face significant credit constraints (Beck et al., 2012) that limit growth (Hutchinson & Xavier, 2006). Private credit also plays an elemental role in sustaining small and medium enterprises that are the drivers of employment and economic growth (Love & Roper, 2015).

Evans and Jovanovic (1989) concluded that liquidity constraints are the most significant barrier to entrepreneurship. They argue that capital is essential for starting a business, and liquidity constraints tend to exclude those with insufficient funds at their disposal. More specifically, they estimate that entrepreneurs are limited to a capital stock that is about one and one-half times their wealth, and that this is often insufficient to provide the needed funds for a successful launch of a new business. In their examination of the influence of inheritance on entrepreneurial growth and success, Holtz-Eakin, Joulfaian, and Rosen (1994) concur with the Evans and Jovanovic thesis. Using an examination of tax returns, they conclude that liquidity constraints constrain entrepreneurial initiatives, whereas the presence of liquidity in the form of inheritance has a discernibly positive influence on the viability and profitability of sole proprietorships. Specifically, they find that a "$150,000 inheritance increases the probability that an individual will continue as a sole proprietor by 1.3 percentage points, and if the enterprise survives, its receipts increase by almost 20 percent" (p. 54). A further refinement of this logic and one that speaks directly to our analysis was provided by Beck and colleagues (2012). They find that greater levels of firm (enterprise) credit are positively associated with economic growth and with greater reductions in income inequality.

Credit availability from the financial sector could be positively related to growth under increasing inequality for the following reasons. First, various forms of external financing constraints could impede the growth of small firms, the engines of economic growth (Freel, 2000). Small firms, due to higher information opacity, less reliable credit history, limited collateral, and higher monitoring costs (Berger & Udell, 2006), may find it harder to borrow. To overcome these lending challenges, government programs increase access to credit by loan guarantee programs through private financial institutions.

Although microloans are a financing mechanism to overcome the above challenges, microloans focus on very small firms managed by the working poor. Broader credit availability from the financial sector signals efficiency in the allocation of capital, lower financing constraints, access to investment opportunities, and improved human capital (Levine, 2005). More credit from financial sector could help loosen the grip of inequality on growth by increasing financing available to exploit opportunities, both for entrepreneurs and small and medium enterprises. When credit is limited, opportunities for experimentation with resources are limited, thereby entrenching more inefficient business models.

Vibrant credit markets are essential for young firms and small to medium enterprises, the engines of economic growth. Lower participation by the private financial sector in lending efforts could increase credit rationing and limit access to credit, further solidifying the strength of association between inequality and growth. The expanse of credit institutions is much broader than that of venture capitalists and angel investors. Early stage survival of a venture is contingent on completing a series of resource acquisition activities that require cash injections. Access to credit provides the fungible resources during the early stages of firm development or for sustaining small ongoing businesses. Faced with limited growth opportunities, domestic credit could help small firms with unstable cash flows to survive and pursue additional business activities.

Second, greater participation by financial institutions in lending activities further weakens the grip of inequality in limiting growth. Inequality increases frictions in the credit markets by allowing a select few to have greater access to economic opportunities. This selective access for a few lowers the economic multipliers that could enhance growth. Greater cronyism in less competitive lending markets primed by economic inequality also has social and political implications for growth. Lower participation in lending from the financial sector could further lower long-term human capital accumulation. With limited credit restricting the loci of opportunities at the economy level, opportunities for developing human capital could be limited; therefore, further allowing a select few with more economic capital to limit growth for their own gains. When the financial sector participates in the economy to a lower extent, economic resilience could decline because of weaker undergirds to a financial system less dependent on private sector vibrancy. A healthy and a competitive lending system in a country could lower grip of stakeholders benefiting from inequality, and improve the perception of fairness and financial inclusion.

Overall, we expect that with increasing income inequality, higher domestic credit from the financial sector will be positively associated with growth.

**Hypothesis 2:** With increasing income inequality, higher domestic credit provided by the financial sector will be positively associated with GDP per capita growth.

## Data, Analysis, and Results

To test the proposed hypotheses, we collated data on inequality and entrepreneurship from a range of publicly available data sources. Below, we provide a description of our sample, estimation techniques, results, and robustness checks.

### *Sample*

The primary sources of our data are the World Development Indicators data, the Penn World Table version 9 (Feenstra, Inklaar, & Timmer, 2015), GeoDist Data from centre d'etudes prospectives et d'informationales internationales in France,[4] and additional country-specific data from Ashraf and Galor (2013). To test for endogeneity, we draw on a country-wise measure of wheat–sugar ratio from Easterly (2007).[5] The wheat–sugar ratio is the ratio of share of arable land suitable for cultivating wheat to the share of arable land suitable for cultivating sugar.

After merging the databases and including a variety of geographic, demographic, and economic controls, our final sample, based on casewise deletion, includes 480 country–year observations from 92 countries. The countries represented in the sample include both developed and developing countries with full information available for the period from 2004 to 2014. Table 2 presents the list of countries and mean values for the key variables of interest.

### *Analytical Specification*

Because several of the control variables are time invariant, to model their effects in a panel data set, we use a random-effects specification. The outcome variable is the percentage GDP per capita growth (annual percentage). This measure is advantageous for two reasons. First, GDP per capita may be influenced by country-specific population factors and baseline GDP levels. Second, defining the dependent variable as the percentage change in GDP per capita helps us capture variation in economic growth that can be influenced by variations in entrepreneurial activity.

As a measure of income inequality, we use the Gini index available from the World Bank's World Development Indicators database.[6] The Gini

**Table 2.** Country-Wise Means.

| Number | Country name | GDP per capita growth (annual %) | Gini index (World Bank estimate)[a] | Ease of startup | Domestic credit provided by the financial sector (% of GDP)[b] |
|---|---|---|---|---|---|
| 1 | Albania | 5.4 | 29.85 | 0.18 | 61.54 |
| 2 | Argentina | 3.48 | 45.98 | −0.43 | 29.53 |
| 3 | Armenia | 6.62 | 32.05 | 0.41 | 23.27 |
| 4 | Australia | 1.04 | 35.28 | 0.94 | 156.72 |
| 5 | Austria | 1.25 | 30.25 | 0.22 | 129.17 |
| 6 | Azerbaijan | 8.47 | 31.79 | 0.55 | 16.22 |
| 7 | Bangladesh | 4.68 | 32.67 | −0.06 | 52.54 |
| 8 | Belarus | 7.03 | 27.69 | −0.16 | 31.05 |
| 9 | Belgium | 0.85 | 28.95 | 0.82 | 109.19 |
| 10 | Bhutan | 9.37 | 38.45 | 0.09 | 37.35 |
| 11 | Bolivia | 3.08 | 52.03 | −0.82 | 53.93 |
| 12 | Bosnia and Herzegovina | 1.82 | 33.83 | −0.69 | 54.46 |
| 13 | Botswana | −9.22 | 60.46 | −0.25 | −1.11 |
| 14 | Brazil | 2.3 | 54.46 | −1.34 | 88.38 |
| 15 | Bulgaria | 3.25 | 33.89 | 0.23 | 59.93 |
| 16 | Burkina Faso | −0.11 | 39.76 | 0.66 | 14.86 |
| 17 | Cambodia | 5.22 | 36.59 | −1.02 | 14.06 |
| 18 | Canada | 1.57 | 33.89 | 0.93 | 169.22 |
| 19 | Chile | 2.59 | 51.54 | 0.15 | 97.71 |
| 20 | Colombia | 2.91 | 55.04 | 0.16 | 61.28 |
| 21 | Costa Rica | 3.14 | 48.94 | −0.56 | 49.81 |
| 22 | Croatia | −0.25 | 31.46 | 0.19 | 79.4 |
| 23 | Czech Republic | 2.45 | 26.54 | 0.09 | 55.8 |
| 24 | Denmark | 0.46 | 27.9 | 0.79 | 216.31 |
| 25 | Dominican Republic | 2.44 | 47.29 | 0.32 | 42.7 |
| 26 | El Salvador | 1.35 | 44.91 | −0.06 | 62 |
| 27 | Estonia | 4.31 | 32.84 | 0.44 | 80.48 |
| 28 | Finland | 0.97 | 27.75 | 0.86 | 126.61 |
| 29 | France | 1.08 | 32.07 | 0.69 | 131.57 |
| 30 | Gabon | 1.02 | 42.18 | −0.06 | 9.77 |
| 31 | Georgia | 7.42 | 40.79 | 0.61 | 32 |
| 32 | Germany | 2.16 | 31.54 | 0.16 | 137.56 |
| 33 | Ghana | 3.17 | 42.77 | −0.14 | 30.84 |

*(continued)*

**Table 2. (continued)**

| Number | Country name | GDP per capita growth (annual %) | Gini index (World Bank estimate)[a] | Ease of startup | Domestic credit provided by the financial sector (% of GDP)[b] |
|---|---|---|---|---|---|
| 34 | Greece | 0.39 | 34.52 | −0.62 | 113.96 |
| 35 | Guatemala | 2.48 | 53.62 | −0.41 | 37.86 |
| 36 | Guinea | 1.67 | 33.73 | 0.2 | 21.58 |
| 37 | Haiti | 1.44 | 60.79 | −1.52 | 21.02 |
| 38 | Hungary | 1.22 | 29.42 | 0.46 | 72.25 |
| 39 | Iceland | 0.69 | 28.78 | 0.7 | 251.15 |
| 40 | India | 5.25 | 35.15 | −0.55 | 76.15 |
| 41 | Ireland | 0.01 | 32.56 | 0.75 | 202.19 |
| 42 | Israel | 3.48 | 41.94 | 0.59 | 87.98 |
| 43 | Italy | −0.54 | 34.17 | 0.32 | 136.04 |
| 44 | Japan | −1.14 | 32.11 | 0.23 | 295.55 |
| 45 | Kazakhstan | 5.6 | 28.99 | 0.15 | 45.81 |
| 46 | Kenya | 3.05 | 48.51 | −0.73 | 37.36 |
| 47 | Kyrgyz Republic | 2.72 | 30.6 | 0.54 | 11.81 |
| 48 | Latvia | 6.81 | 36.5 | 0.65 | 75.64 |
| 49 | Lesotho | 5.42 | 54.18 | 0.19 | −5.71 |
| 50 | Lithuania | 7.95 | 35.08 | 0.29 | 49.37 |
| 51 | Luxembourg | 0.22 | 32.01 | 0.57 | 190.69 |
| 52 | Macedonia, FYR | 5.18 | 43.34 | 0.13 | 31.34 |
| 53 | Madagascar | 0.25 | 42.65 | 0.24 | 13.48 |
| 54 | Malawi | 2.6 | 39.87 | −0.33 | 12.12 |
| 55 | Malaysia | 2.63 | 46.12 | −0.18 | 122.65 |
| 56 | Mexico | 2.09 | 48.26 | 0.26 | 39.14 |
| 57 | Moldova | 4.31 | 34.79 | −0.04 | 36.56 |
| 58 | Mongolia | 5.86 | 32.04 | 0.66 | 66.05 |
| 59 | Morocco | 6.33 | 40.72 | 0.52 | 74.29 |
| 60 | Namibia | −1.14 | 60.97 | −0.33 | 44.96 |
| 61 | Nepal | 3.72 | 32.84 | 0.21 | 67.35 |
| 62 | Netherlands | 0.9 | 29.43 | 0.55 | 193.47 |
| 63 | Niger | 0.05 | 40.86 | −0.8 | 8.04 |
| 64 | Nigeria | 4.13 | 42.97 | 0.1 | 37.11 |
| 65 | Norway | 0.6 | 27.8 | 0.69 | 116.77 |
| 66 | Pakistan | 2.7 | 31.56 | −0.3 | 44.77 |
| 67 | Panama | 6.12 | 53.04 | 0.51 | 82.31 |

*(continued)*

**Table 2. (continued)**

| Number | Country name | GDP per capita growth (annual %) | Gini index (World Bank estimate)[a] | Ease of startup | Domestic credit provided by the financial sector (% of GDP)[b] |
|---|---|---|---|---|---|
| 68 | Peru | 4.74 | 48.35 | −0.09 | 18.76 |
| 69 | Philippines | 2.62 | 43.38 | −0.93 | 49.29 |
| 70 | Poland | 4.77 | 34 | −0.09 | 50.84 |
| 71 | Portugal | 0.66 | 37.08 | 0.25 | 167.15 |
| 72 | Russian Federation | 4.56 | 41.2 | 0.18 | 29.45 |
| 73 | Rwanda | 5.88 | 51.69 | 0.19 | 7.98 |
| 74 | Senegal | 0.83 | 39.76 | 0.08 | 26.47 |
| 75 | Sierra Leone | 2.39 | 33.99 | 0.38 | 16.35 |
| 76 | Slovak Republic | 4.48 | 26.98 | 0.3 | 55.95 |
| 77 | Slovenia | 1.4 | 24.7 | 0.21 | 75.19 |
| 78 | South Africa | 2.6 | 63.73 | 0.16 | 177.33 |
| 79 | Spain | −0.22 | 34.49 | −0.31 | 207.07 |
| 80 | Sri Lanka | 5.54 | 37.77 | 0.04 | 46.6 |
| 81 | Sweden | 1.3 | 26.89 | 0.84 | 136.91 |
| 82 | Switzerland | 0.57 | 32.93 | 0.49 | 173.73 |
| 83 | Tajikistan | 4.05 | 31.16 | −0.16 | 16.65 |
| 84 | Thailand | 3.6 | 40.03 | 0.13 | 128.66 |
| 85 | Togo | 1.71 | 44.11 | −0.72 | 26.26 |
| 86 | Tunisia | 2.49 | 36.77 | 0.2 | 68.83 |
| 87 | Turkey | 4.29 | 39.89 | 0.44 | 54.54 |
| 88 | Uganda | 2.07 | 42.72 | −0.85 | 10.3 |
| 89 | Ukraine | 2.43 | 26.49 | −0.04 | 76 |
| 90 | United Kingdom | 0.41 | 34.63 | 0.55 | 180.2 |
| 91 | Uruguay | 5.42 | 45.6 | −0.13 | 33.47 |
| 92 | Zambia | 5.46 | 54.84 | 0.24 | 19.92 |
| | Average | **2.83** | **36.91** | **0.17** | **87.05** |

*Source*. Click on the tab "Details" at https://data.worldbank.org/indicator/FS.AST.DOMS.GD.ZS

[a]The mean values based on casewise deletions based on variables of the full model in Table 4.

[b]Some values can be negative. The explanation for the negative value is "In a few countries governments may hold international reserves as deposits in the banking system rather than in the central bank. Because claims on the central government are a net item (claims on the central government minus central government deposits), the figure may be negative, resulting in a negative figure for domestic credit provided by the banking sector."

coefficient, a measure of income inequality ranging from 0 to 1, measures the distribution of income at the individual or at household levels relative to a hypothetical distribution in which every individual or household had equal income. A Gini coefficient of 0 indicates that income is uniformly distributed in a population, whereas an increasing value suggests increasing inequality. A value of 1, indicating the highest level of inequality possible, indicates that one person has all the income in a country.

For the ease of startup measure, we create a factor score of three indicators: (a) startup procedures to register a business, female (number);[7] (b) time required to start a business (days); and (c) cost of business startup procedures (% of Gini per capita). Because cultural, religious, and institutional factors may impose more restrictions on females, the first indicator refers to procedures necessary for females. The two latter measures focus on resource requirements to start a business. The factor score is multiplied by −1, so higher values indicate greater ease of startup.

Domestic credit to the private sector is measured as financial lending from financial institutions to the private sector as a percentage of GDP and is based on World Bank's World Development Indicators database.[8] Private sector financial support includes loans, purchase of nonequity securities, trade credit, and loans against claims of repayments. These financial corporations include private and public institutions and depository banks, including insurance firms, pension funds, and foreign exchange companies.

We include a variety of controls related to geographic, demographic, and economic factors. To control for the overall entrepreneurial climate at the country level, we use the standard error of the prediction resulting from regressing rate of new entry on ease of startup and domestic credit to private sector as predictors (specifically, using *-predict, stdp-* command in Stata 15). This approach also lowers multicollinearity between ease of startup, domestic credit to private sector, and rate of new entry. The rate of new entry is the number of new business registrations (number of new limited liability corporations) in a calendar year per 1,000 individuals between the ages 15 and 64 years. This measure is collated from the World Bank's World Development Indicators database[9] and, based on the guidelines from the World Bank, it is labeled as firm density. We take the natural log of this measure to lower the influence of extreme values.

We include population (in log terms) as a control variable because countries with a larger population are likely to grow at a faster rate (Becker, Glaeser, & Murphy, 1999). Next, we include the log of the geographic area to control for land mass (Olsson & Hibbs, 2005). As landlocked countries may lack direct import and export opportunities through sea routes, we include whether a country is landlocked as a control (0 = not landlocked, 1 = landlocked).

The size of the country and socioeconomic development could be strongly correlated with the number of cities and the number of cities is important to economic growth (Hoselitz, 1953). Furthermore, as the rate of urbanization increases with the number of cities, income inequality first increases, and then declines as urbanization increases over time. This nonlinear relationship is also known as Kuznets inverted-U. To lower confound between inequality and urbanization, we control for the number of cities.

As ethnic fractionalization has been shown to influence cultural, social, and political outcomes, we include this measure as a control (Fearon, 2003). We also include life expectancy at birth as a control because it proxies for future life opportunities and health infrastructure.

Next, we control for a series of economic factors. To control for the size of economy, we include the log of GDP adjusted for purchasing power parity in current international dollars. As returns from natural resources could influence industrial activity, we include total natural resource rents (% of GDP). As the size of the active labor force directly influences economic activity, we also include the log of the total labor force.

We include a share of household consumption adjusted for purchasing power parity. Based on Penn World Tables version 8.0, this measure is the share of output-based real GDP per capita represented in household consumption at current prices, adjusted for purchasing power parity. In other words, it is the consumption share of purchasing power parity converted GDP per capita at current prices.

To control for trade activity that could influence firm turnover, we control for trade as a percentage of GDP. To control for agricultural opportunities, we include as control a measure of land suitability for agriculture from Ashraf and Galor (2013). Finally, to control for capital accumulation, we include gross capital formation as a percentage of GDP; and to control for the entrepreneurial impetus provided by foreign direct investments, we include new inflows of foreign investments as a percentage of GDP.

Table 2 shows the country-specific mean of key predictors. The highest Gini coefficient is in South Africa (63.73) and the lowest is in Slovenia (24.70). The mean Gini coefficient is 36.91. Australia and Canada are among the top countries in the ease of startup, whereas Brazil and Haiti are among the bottom. Japan and Iceland have the highest domestic credit from financial sector, whereas Botswana and Lesotho have the lowest.

**Table 3.** Sample Descriptive Statistics.

| | | Source | *M* | *SD* | Minimum | Maximum | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | GDP per capita growth (annual %) | WDI | 2.83 | 4.22 | −14.56 | 15.40 | 1 | | | | | | | | | | | | | | | | | |
| 2 | Gini index | WDI | 36.91 | 8.97 | 23.72 | 64.79 | .13 | 1 | | | | | | | | | | | | | | | | |
| 3 | Ease of startup | WDI | 0.17 | 0.55 | −1.75 | 0.94 | −.17 | −.46 | 1 | | | | | | | | | | | | | | | |
| 4 | Domestic credit provided by financial sector (% of GDP) | WDI | 87.05 | 63.87 | −5.71 | 301.17 | −.4 | −.32 | .39 | 1 | | | | | | | | | | | | | | |
| 5 | Log of new business density (residual after using moderators as predictors) | WDI | 0.06 | 0.01 | 0.05 | 0.13 | −.22 | −.09 | .02 | .67 | 1 | | | | | | | | | | | | | |
| 6 | Log of population | Penn World Tables v.9 | 2.63 | 1.22 | 0.26 | 7.13 | −.06 | .24 | −.29 | −.03 | .03 | 1 | | | | | | | | | | | | |
| 7 | Log of geographic area | CEPII | 12.18 | 1.65 | 7.86 | 16.65 | .02 | .32 | −.28 | −.15 | .03 | .73 | 1 | | | | | | | | | | | |
| 8 | Landlocked | CEPII | 0.23 | 0.42 | 0 | 1 | .05 | −.14 | −.01 | −.21 | −.15 | −.25 | −.16 | 1 | | | | | | | | | | |
| 9 | Number of cities | CEPII | 24.89 | 1.10 | 11 | 25 | −.09 | −.14 | .02 | .12 | .02 | .08 | .11 | −.17 | 1 | | | | | | | | | |
| 10 | Ethnic fractionalization | Ashraf and Galor (2013) | 0.35 | 0.22 | 0.01 | 0.93 | .15 | .41 | −.26 | −.4 | −.16 | .11 | .22 | .2 | 0 | 1 | | | | | | | | |
| 11 | Life expectancy at birth, total (years) | WDI | 74.11 | 6.75 | 46.96 | 82.92 | −.24 | −.39 | .41 | .61 | .25 | −.13 | −.23 | −.25 | .25 | −.61 | 1 | | | | | | | |
| 12 | Log of GDP adjusted for PPP in current international dollars | WDI | 25.89 | 1.61 | 21.98 | 29.39 | −.21 | −.04 | .02 | .36 | .16 | .81 | .56 | −.31 | .2 | −.23 | .37 | 1 | | | | | | |
| 13 | Total natural resources rents (% of GDP) | WDI | 3.75 | 5.96 | 0 | 41.19 | .12 | .23 | −.15 | −.33 | −.13 | .18 | .51 | .23 | −.02 | .4 | −.51 | −.02 | 1 | | | | | |
| 14 | Log of total labor force | WDI | 15.54 | 1.37 | 12.07 | 19.98 | −.06 | .22 | −.28 | −.03 | .03 | .99 | .73 | −.24 | .08 | .09 | −.1 | .83 | .19 | 1 | | | | |
| 15 | Share of household consumption at current PPPs | Penn World Tables v.9 | 0.62 | 0.13 | 0.11 | 1.02 | .15 | .36 | −.37 | −.46 | −.14 | .01 | −.11 | .02 | −.11 | .2 | −.42 | −.37 | −.09 | −.02 | 1 | | | |
| 16 | Trade (% of GDP) | WDI | 91.16 | 45.87 | 22.11 | 343.56 | .02 | −.35 | .29 | .19 | −.12 | −.51 | −.57 | .27 | .02 | .05 | .14 | −.3 | −.19 | −.51 | −.27 | 1 | | |
| 17 | Land suitability Gini | Ashraf and Galor (2013) | 0.35 | 0.23 | 0.03 | 0.87 | −.08 | −.01 | .16 | .09 | .09 | .18 | .49 | 0 | .06 | .03 | .04 | .19 | .31 | .18 | −.29 | −.22 | 1 | |
| 18 | Gross capital formation (% of GDP) | WDI | 0.23 | 0.07 | 0.06 | 0.56 | −.02 | −.2 | .21 | .46 | .31 | −.22 | −.24 | −.03 | .01 | −.17 | .41 | .1 | −.27 | −.23 | −.44 | .29 | −.09 | 1 |
| 19 | Foreign direct investment, net inflows (% of GDP) | WDI | 6.43 | 14.79 | −58.32 | 252.31 | −.02 | −.09 | .13 | .14 | −.04 | −.18 | −.25 | .11 | .03 | .04 | .09 | −.08 | −.04 | −.19 | −.12 | .43 | −.05 | .2 |

*Source.* http://databank.worldbank.org/data/reports.aspx?source=world-development-indicators. CEPII = GeoDist data from http://www.cepii.fr/CEPII/en/bdd_modele/presentation.asp?id=6] Mayer and Zignago (2011). Penn World Tables v.9 (https://www.rug.nl/ggdc/productivity/pwt/) Feenstra, Inklaar, and Timmer (2015).
*Note.* $N = 480$ country–year observations from 92 countries; casewise deletions based on variables the full model in Table 4; correlations $\geqslant .|09|$ are significant at $p <$ .05 (two tailed). CEPII = centre d'etudes prospectives et d'informations internationales; PPP = purchasing power parity; WDI = World Development Indicators.

Table 3 presents the data source for each variable, descriptives, and pair-wise correlations.

## Results

The panel represents an unbalanced data set of country–year observations. Because several of our variables are country-specific characteristics, which do not vary over time, we use a random-effects model (*xtreg* routine in Stata 15). To lower concerns for specification bias, we test our results using three different sets of stepwise regressions. In Table 4, we first specify the models without any controls and no casewise deletion (Models 1-4). Models with controls but no casewise deletion are presented next (Models 5-8), followed by models with controls and casewise deletion (Models 8-12). Our findings are generally consistent across all the models.

Hypothesis 1 proposed that with increasing income inequality, ease of startup will be positively associated with GDP per capita growth (Table 4, Model 10: $\beta = 0.181$, $p < .01$). Figure 1 shows that with increasing Gini index, greater ease of startup is positively associated with GDP per capita growth. The effects are that for a one-unit increase in the ease of startup and a one-unit increase in the Gini index, annual GDP per capita growth increases by 0.91%.[10]

Hypothesis 2 proposed that with increasing income inequality, the availability of credit from the private sector would be positively associated with GDP per capita growth. This is substantiated in Table 4 (Model 11: $\beta = 0.00158$, $p < .01$) with the direction of effects consistent with the proposed hypotheses. In Figure 2, higher levels of availability of domestic credit from the financial sector are positively associated with GDP per capita growth. The effects are that for a 10-unit increase in access to credit and a 10-unit increase in the Gini index, annual GDP per capita growth increases by 0.08%.

Overall, with increasing income inequality, ease of startup (Hypothesis 1) or domestic credit from financial institutions (Hypothesis 2) are positively associated with GDP per capita growth.

## Robustness Checks

Next, we test whether the results are robust to alternate specifications.

*Autoregressive (AR) specification.* We specify an AR(1) specification using *xtregar* routine in Stata 15. The estimates in Table 5 (Model 1) are consistent with the main model.

**Table 4.** Random-Effects Estimates.

| | GDP per capita growth (annual %) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | No controls | | | | No casewise deletion | | | | Casewise deletion | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| Ease of startup | | −9.512*** | | −6.316*** | | −8.571*** | | −5.468*** | | −8.685*** | | −5.554*** |
| | | (1.262) | | (1.330) | | (1.393) | | (1.436) | | (1.754) | | (1.779) |
| Gini index (World Bank estimate) | | −0.0206 | −0.150*** | −0.114*** | | −0.00383 | −0.124*** | −0.0895** | | 0.00463 | −0.0941** | −0.0732 |
| | | (0.0252) | (0.0217) | (0.0376) | | (0.0302) | (0.0365) | (0.0405) | | (0.0360) | (0.0472) | (0.0497) |
| Ease of startup × Gini index (Hypothesis 1) | | 0.192*** | | 0.130*** | | 0.166*** | | 0.113*** | | 0.181*** | | 0.121*** |
| | | (0.0310) | | (0.0322) | | (0.0328) | | (0.0335) | | (0.0424) | | (0.0422) |
| Domestic credit provided by the financial sector (% of GDP) | | | −0.0709*** | −0.0594*** | | | −0.109*** | −0.0842*** | | | −0.0903*** | −0.0714*** |
| | | | (0.0105) | (0.0163) | | | (0.0164) | (0.0177) | | | (0.0181) | (0.0191) |
| Domestic credit × Gini index (Hypothesis 2) | | | 0.00137*** | 0.00113** | | | 0.00181*** | 0.00131*** | | | 0.00158*** | 0.00120** |
| | | | (0.000288) | (0.000449) | | | (0.000438) | (0.000471) | | | (0.000500) | (0.000519) |
| Log of new business density (residual after using moderators as predictors) | | | | | −20.44 | −98.30*** | 78.50** | 48.83 | −108.0*** | −109.9*** | 27.81 | 5.339 |
| | | | | | (18.92) | (27.98) | (32.90) | (35.48) | (29.70) | (29.36) | (37.39) | (38.77) |
| Log of population | | | | | 2.794*** | −0.986 | −0.0177 | −0.414 | −0.737 | −1.271 | 0.0750 | −0.362 |
| | | | | | (0.997) | (1.409) | (1.247) | (1.259) | (1.547) | (1.528) | (1.373) | (1.387) |
| Log of the geographic area | | | | | 0.328 | 0.234 | 0.0494 | 0.0757 | 0.467 | 0.342 | 0.191 | 0.198 |
| | | | | | (0.239) | (0.302) | (0.269) | (0.270) | (0.327) | (0.329) | (0.302) | (0.303) |
| Landlocked | | | | | 0.0603 | −0.296 | −0.746 | −0.647 | −0.960 | −0.817 | −1.092* | −0.968 |
| | | | | | (0.559) | (0.625) | (0.552) | (0.553) | (0.704) | (0.711) | (0.629) | (0.636) |
| Number of cities | | | | | 0.160** | 0.00614 | −0.0656 | −0.0755 | −0.0899 | −0.0936 | −0.195 | −0.166 |
| | | | | | (0.0629) | (0.152) | (0.141) | (0.141) | (0.196) | (0.200) | (0.188) | (0.192) |
| Ethnic fractionalization | | | | | −3.413*** | −2.393* | −3.097*** | −2.883** | −2.374 | −2.727* | −3.047** | −3.083** |
| | | | | | (1.028) | (1.330) | (1.180) | (1.187) | (1.530) | (1.592) | (1.418) | (1.443) |
| Life expectancy at birth, total (years) | | | | | 0.0608 | 0.0150 | 0.0361 | 0.0352 | 0.0576 | 0.0521 | 0.0884 | 0.0758 |
| | | | | | (0.0382) | (0.0551) | (0.0502) | (0.0501) | (0.0734) | (0.0723) | (0.0671) | (0.0673) |
| Log of GDP adjusted for PPP in current international dollars | | | | | −1.771*** | −0.873** | −0.378 | −0.186 | −2.277*** | −1.669*** | −1.335** | −1.138** |
| | | | | | (0.319) | (0.436) | (0.416) | (0.418) | (0.562) | (0.576) | (0.562) | (0.569) |

*(continued)*

**Table 4. (continued)**

| | GDP per capita growth (annual %) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | No controls | | | | No casewise deletion | | | | Casewise deletion | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| Total natural resources rents (% of GDP) | | | | | 0.0814*** | 0.0252 | 0.0204 | 0.00967 | 0.0698 | 0.0706 | 0.0542 | 0.0515 |
| | | | | | (0.0180) | (0.0317) | (0.0295) | (0.0295) | (0.0534) | (0.0527) | (0.0497) | (0.0500) |
| Log of total labor force | | | | | −0.634 | 1.634 | 0.613 | 0.654 | 2.812** | 2.548* | 1.279 | 1.361 |
| | | | | | (0.874) | (1.222) | (1.096) | (1.098) | (1.390) | (1.371) | (1.257) | (1.260) |
| Share of household consumption at current PPPs | | | | | −1.152 | −0.241 | 0.632 | 0.0632 | 1.662 | 1.133 | −0.342 | −0.238 |
| | | | | | (1.410) | (2.048) | (1.925) | (1.922) | (2.658) | (2.649) | (2.514) | (2.508) |
| Trade (% of GDP) | | | | | 0.0178*** | 0.0133** | 0.0215*** | 0.0211*** | 0.0170** | 0.0180** | 0.0198*** | 0.0197*** |
| | | | | | (0.00517) | (0.00643) | (0.00609) | (0.00613) | (0.00797) | (0.00794) | (0.00734) | (0.00738) |
| Land suitability Gini | | | | | −2.141* | 0.826 | 0.658 | 1.488 | −0.720 | 1.182 | 0.303 | 1.234 |
| | | | | | (1.262) | (1.443) | (1.260) | (1.278) | (1.449) | (1.479) | (1.287) | (1.327) |
| Gross capital formation (% of GDP) | | | | | 8.990*** | 16.11*** | 16.26*** | 14.68*** | 22.31*** | 19.97*** | 19.48*** | 18.38*** |
| | | | | | (1.924) | (2.969) | (2.780) | (2.817) | (3.794) | (3.742) | (3.609) | (3.613) |
| Foreign direct investment, net inflows (% of GDP) | | | | | 0.00514 | −0.00653 | −0.00245 | −0.00441 | −0.00909 | −0.00873 | −0.00524 | −0.00555 |
| | | | | | (0.0121) | (0.0125) | (0.0124) | (0.0123) | (0.0135) | (0.0131) | (0.0131) | (0.0130) |
| Constant | 2.130*** | 4.259*** | 9.980*** | 9.072*** | 37.86*** | 0.706 | 0.162 | −3.698 | 11.87 | 4.521 | 13.11 | 8.372 |
| | (0.165) | (0.987) | (0.893) | (1.466) | (11.29) | (16.76) | (15.41) | (15.42) | (18.84) | (18.69) | (17.41) | (17.50) |
| Observations | 4,792 | 751 | 1,107 | 741 | 1,591 | 680 | 680 | 680 | 480 | 480 | 480 | 480 |
| Number of countries (ISO3N) | 204 | 144 | 148 | 141 | 140 | 119 | 119 | 119 | 92 | 92 | 92 | 92 |
| Chi-square | | 75.77 | 115.8 | 118.6 | 118.6 | 131.1 | 162.3 | 180.9 | 70.05 | 101.1 | 117.1 | 128.6 |
| *df* | | 3 | 3 | 5 | 15 | 18 | 18 | 20 | 15 | 18 | 18 | 20 |
| *p* | | <.001 | <.001 | <.001 | <.001 | <.001 | <.001 | <.001 | <.001 | <.001 | <.001 | <.001 |

*Note*. Standard errors in parentheses. PPP = purchasing power parity.
$^{*}p < .1$. $^{**}p < .05$. $^{***}p < .01$ (two tailed).

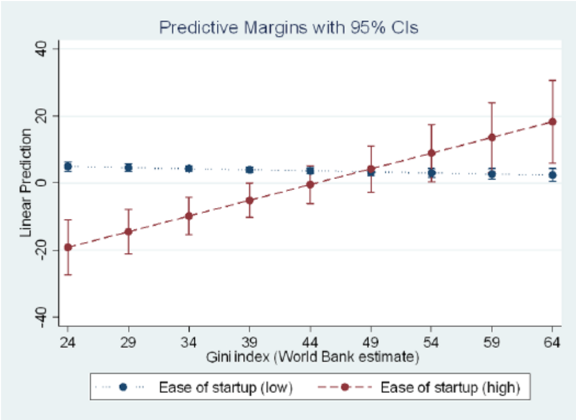


**Figure 1.** Moderation effect of ease of startup.
*Note.* CI = confidence interval.

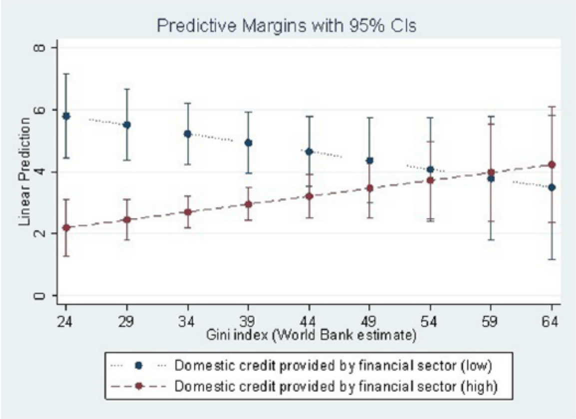


**Figure 2.** Moderation effect of domestic credit.
*Note.* CI = confidence interval.

*Endogeneity.* It is possible that inequality and growth per capita are endogenous. Based on Easterly (2007), we use as an instrument the ratio of agricultural land suitable for cultivating wheat to agricultural land suitable for growing sugarcane to explore this possibility. Suitability of land for growing either crop is based on

**Table 5.** Robustness Tests: Autocorrelation and 2SLS Estimates.

| | (1) | (2) | (3) |
|---|---|---|---|
| | *Xtregar*—main model | First stage | Second stage |
| Variables | DV = GDP per capita growth (annual %) | DV = Gini index | DV = GDP per capita growth (annual %) |
| Ease of startup | −4.788** | | −5.554*** |
| | (1.889) | | (1.779) |
| Gini index | −0.0729 | | −0.0732 |
| | (0.0479) | | (0.0497) |
| Ease of startup × Gini index (Hypothesis 1) | 0.111** | | 0.121*** |
| | (0.0439) | | (0.0422) |
| Domestic credit provided by the financial sector (% of GDP) | −0.0669*** | | −0.0714*** |
| | (0.0188) | | (0.0191) |
| Domestic credit × Gini index (Hypothesis 2) | 0.00115** | | 0.00120** |
| | (0.000499) | | (0.000519) |
| Wheat-to-sugarcane ratio | | −27.41*** | |
| | | (7.230) | |
| Wheat-to-sugarcane ratio × Wheat-to-sugarcane ratio | | 49.20*** | |
| | | (18.99) | |
| Log of new business density (residual after using moderators as predictors) | 1.352 | 37.66 | 5.339 |
| | (39.77) | (31.99) | (38.77) |
| Log of population | −0.642 | −0.475 | −0.362 |
| | (1.336) | (2.824) | (1.387) |
| Log of geographic area | 0.240 | 2.501*** | 0.198 |
| | (0.290) | (0.820) | (0.303) |
| Landlocked | −0.799 | −4.957*** | −0.968 |
| | (0.602) | (1.888) | (0.636) |
| Number of cities | −0.184 | −0.158 | −0.166 |
| | (0.185) | (0.379) | (0.192) |
| Ethnic fractionalization | −2.719** | −9.366*** | −3.083** |
| | (1.375) | (3.500) | (1.443) |

*(continued)*

**Table 5. (continued)**

| Variables | (1) *Xtregar*—main model DV = GDP per capita growth (annual %) | (2) First stage DV = Gini index | (3) Second stage DV = GDP per capita growth (annual %) |
|---|---|---|---|
| Life expectancy at birth, total (years) | 0.0739 | −0.513*** | 0.0758 |
| | (0.0646) | (0.0963) | (0.0673) |
| Log of GDP adjusted for PPP in current international dollars | −1.082* | −1.365** | −1.138** |
| | (0.554) | (0.622) | (0.569) |
| Total natural resources rents (% of GDP) | 0.0577 | −0.0706 | 0.0515 |
| | (0.0492) | (0.0444) | (0.0500) |
| Log of total labor force | 1.555 | −0.0666 | 1.361 |
| | (1.220) | (2.567) | (1.260) |
| Share of household consumption at current PPPs | 0.561 | −5.802*** | −0.238 |
| | (2.496) | (2.162) | (2.508) |
| Trade (% of GDP) | 0.0190*** | 0.0152* | 0.0197*** |
| | (0.00724) | (0.00830) | (0.00738) |
| Land suitability Gini | 0.836 | −4.347 | 1.234 |
| | (1.269) | (4.557) | (1.327) |
| Gross capital formation (% of GDP) | 16.99*** | −7.199*** | 18.38*** |
| | (3.766) | (2.572) | (3.613) |
| Foreign direct investment, net inflows (% of GDP) | −0.00880 | −0.0402** | −0.00555 |
| | (0.0134) | (0.0157) | (0.0130) |
| Constant | 4.451 | 94.95*** | 8.372 |
| | (16.92) | (31.48) | (17.50) |
| Observations | 480 | 615 | 480 |
| Number of countries (ISO3N) | 92 | 98 | 92 |
| Chi-square | 104.8 | 209.3 | 128.6 |
| *df* | 21 | 17 | 20 |
| *p* | $<.001$ | $<.001$ | $<.001$ |

*Note.* Standard errors in parentheses. 2SLS = two-stage least squares; DV = dependent variable; PPP = purchasing power parity.
$^{*}p < .1$. $^{**}p < .05$. $^{***}p < .01$ (two tailed).

climate, soil, and topography, factors exogenous to the proposed relationships. Easterly (2007), drawing on Engerman and Sokoloff (1997), proposed that agricultural endowments could affect inequality. Specifically, the ratio of land suitable for growing wheat relative to land suitable for growing sugarcane predicts the level of inequality. In countries where land was suitable for growing sugarcane, slavery and concentration of wealth emerged due to economies of scale in the growth and trade of sugarcane. Conversely, in countries where agricultural factor endowments were suitable for growing wheat, a middle class emerged as wheat, unlike sugarcane, was generally cultivated in family farms. The instrument of agricultural endowment is exogenous to firm establishment and lending by financial institutions, and is based on climatic and geological conditions instead of active policy efforts to increase economic growth.

We use the linear and squared term of the instrument, controls, and the predictors in the first stage to derive the predicted value of Gini index. In the second stage, we use the predicted Gini measure, instead of the Gini index measure used in the main analysis. All other effects are consistent in direction and significance. To test for the validity of the instrument, due to the short and unbalanced panel along with a non–time-varying instrument, we use the *ivreg2* routine in Stata 15 and control for year and country dummies. The Sargan test statistic was not rejected, suggesting that the instruments used are uncorrelated with the error term and that the excluded instruments are correctly excluded from the estimated equation.

*Recession effects.* Our main analysis controls for the log of total GDP and a log of population in a time series setting to include the effects of year-to-year economic changes, that could include broader events such as a recession or country-specific crisis. Although a recession dummy could model for recessionary effects, such a dummy does not capture the *extent* to which economic activity may go down during a recession. In practice, different countries experience different levels of decline during a recession. Also, some economies may contract even outside of a recession.

To control for recession effects, we include change in GDP as an additional control. The estimates we obtain after including this control are consistent with the main effects in Table 4. Full results are available from the authors on request.

## Discussion and Conclusion

The value of relaxing institutional constraints that restrict entrepreneurship has been discussed in Schumpeter (1934) who argued that economic dynamism comes about through a series of incentives or constraints that can be best understood as an evolutionary process of continuous innovation and "creative destruction." In this study, we explored the potential moderating impact of institutional constraints for creative destruction on the relationship between inequality and growth per capita. We find that under increasing inequality, ease of startup and domestic credit from the financial sector are positively associated with growth per capita. We acknowledge that the effect sizes are small, especially for access to capital. Smaller effect sizes in macro-economic data are expected. A complex confluence of economic, social, institutional, and political factors could slow, dilute, and/or deflate the proposed effects. Smaller effects are also expected due to controls for a variety of factors included in our empirical models. Although our analysis is not definitive, it does provide substantial evidence that with increasing income inequality, policies promoting entrepreneurial activity can have a meaningful association with income growth.

In this final section, we discuss implications of these findings for scholarship and practice, notably for public policy, as well as limitations and suggestions for future research.

### *Theoretical Implications*

Our perspective of institutional constraints is rooted in North (1990), who along with contemporary economists such as Rodrik (2004), argues that cross-country differences in long-term economic performance are attributable to differences in institutional quality. Specifically, North highlighted the importance of a well-functioning legal system, a clearly defined and impartial third (judicial) branch of government to enforce property rights and adjudicate disputes, and a related set of attitudes toward contracting and trading that encourage people to engage in transactions at low costs (North, 1986, 1993). Too many regulations could push prospective entrepreneurs into the informal sector (Baumol, 1990), and even among developed countries, there are systematic differences in the ease of business startup. More stringent entry regulations are known to reduce entry of new firms (Bjørnskov & Foss, 2008; Ciccone & Papaioannou, 2007; Dreher & Gassebner, 2013; Klapper et al., 2006). Our findings add to this body of literature by confirming the economically enabling role of entrepreneurship by reducing institutional constraints on economic growth.

In the broader literature, the potential connection between inequality and growth and development has been a vexing research subject for decades. This research has been both mixed and contradictory; more specifically, a negative relationship between inequality and growth per capita is more likely at lower levels of development and a neutral or positive relationship at higher levels (Kaldor, 1955; Kuznets, 1955). Beyond the level of development, scholars have also explored some contingencies in the relationship; however, to the best of our knowledge, ours is the first study that introduces the role of constraints to entrepreneurship into the equation. The logic behind this focus is that relaxing or increasing institutional constraints has the potential to influence growth per capita, even in the face of structural factors that lead to inequality. As presented in Figures 1 and 2, despite increasing income inequality, lower constraints are shown to promote growth. In our view, this is an important and consequential contribution to the literature and one that provides new directions for understanding—and addressing—the negative consequences of inequality.

Related to the implications for organizational literature on inequality, studies have also focused on demographic inequality, or disparities in endowments, rewards, and experiences based on demographic characteristics, including, but not limited to, age, gender, and race (Wingfield & Taylor, 2016). Unequal treatment of people based on these characteristics generally leads to negative social, psychophysiological, and economic outcomes. Tying these areas in organizational and entrepreneurship research with the current findings, institutional factors that promote participation into self-employment from racial minorities, females, and sexual minorities could further promote growth even in an environment of increasing inequality.

Shifting the level of analysis, some of the recent works in organizational research have also focused on the influence of organizations on inequality and the influence of inequality on organizational outcomes. Powerful organizations can wield substantial influence on changing institutional policies that could increase or decrease inequality (Bapuji, 2015; Cobb, 2016). Economic inequality deteriorates service delivery, lowers corporate reputation, and lowers subsidiary performance (Andrews & Htun, 2018). Soylu and Sheehy-Skeffington (2015) find that social inequality permeates employee behaviors in organizations, in that, "the disproportionate mistreatment of members of low-status groups . . . (has) the intended effect of enhancing the subordination of that group in society at large" (p. 1099). Bapuji (2015) proposes two pathways by which economic inequality influences organizational performance: First, economic inequality increases costs (e.g., higher health care costs, absenteeism) by constraining the development of employee human capital; and second, it negatively affects employee interactions and their contributions to the organizations. Future research may also tie how institutional constraints exacerbate the influence of inequality on venture organization, availability of employees for ventures, and early challenges to venture management. Whether ease of startup and access to credit lower the effects of demographic inequality to promote inclusive innovation (George, McGahan, & Prabhu, 2012) is also an area of future research.

### *Implications for Practice and Policy*

Our research has implications for practice, especially for those engaged in or empowered with developing public policies related to inequality, growth, and development. There is a growing focus on lowering the negative effects of inequality in the policy domain given that inequality can stifle inclusive growth. Indeed, the Ford Foundation, with assets of more than US$12 billion, announced in June 2015 it was concentrating all its US$500 million annual grants on combating inequality.

In recent years, policy makers have focused on policies that "strengthen the rules of the game" to enhance entrepreneurship rates (Sobel, Clark, & Lee, 2007). Government policies toward enhancing inputs include an increase in the quality and quantity of resources. Availability of credit, developing incubators, providing tax reliefs and tax breaks, or providing subsidies are some examples of inputs to priming entrepreneurial activity. Our findings inform policy makers on the importance of supporting the formation of new business, as well as providing access to capital that can support fledgling new businesses and, in so doing, support growth and the broader social and economic development it engenders. Such an approach, although not addressing inequality directly, has the potential to limit its potentially damaging effects on overall economic growth and development. Finally, in terms of returns from such policies, one must keep in mind the small effect sizes. Although the effects of ease of startup procedures are considerable, the effects of access to domestic credit are quite small.

Our research might also have some relevance for global foundations and international institutions seeking to better understand the antecedents and consequences of inequality and development, and other intervening variables and interactions that may influence these two issues independently and in consort.

### *Limitations and Future Research Directions*

Our research has many limitations. First, we focus on one widely used measure of economic growth and development, namely, growth in GDP per capita. There are obviously many other measures and indicators that capture various aspects of economic well-being. However, growth per capita is an inclusive measure that directly relates to economic well-being through higher labor productivity and earnings. It may also be the case that types of entrepreneurship—necessity versus opportunity—could be driving these effects. Second, the purpose of our study is to draw on a large sample of countries, and although we control for a variety of country-level factors, we do not focus on country groups—developing versus developed, transition economies; European Union versus Brazil, Russia, India, China, and South Africa (BRICS), among others. Systematic differences across such clusters of countries could provide additional contingency factors for further exploring the proposed hypotheses.

Third, due to the lack of a clear instrumental variable, we are unable to fully parse out the self-selection effects in which the level of inequality could be a condition of the selection into entrepreneurship. We call on future research to further assess these mechanisms through instruments that influence the relationship between inequality and entrepreneurship association but do not directly affect GDP per capita growth.

Fourth, as is typical of research in this domain, we use specific proxies for our measure of entrepreneurship. The Global Entrepreneurship Monitor (GEM) data has proposed four specific types of total entrepreneurial activity (TEA) entrepreneurship—high growth potential TEA, necessity TEA, opportunity TEA, and overall TEA (Reynolds, Bygrave, Autio, & Hay, 2002). One study found that only high growth potential entrepreneurship had a significant impact on economic growth (Wong, Ho, & Autio, 2005), and so, it may be the case that a finer grained set of entrepreneurship measures would yield different results. Analysis of these outcomes from the GEM data using the current sample did not yield significant results, possibly due to fewer country–year observations represented in the combined sample. GEM focuses on select countries and, therefore, we call for future studies to examine the proposed relationships by drawing on additional characteristics from the GEM data set.

## Conclusion

Our research has been inspired by the call for additional scholarship at the intersection of entrepreneurship, economic development, and institutional quality. We proposed the value of relaxing institutional constraints to entrepreneurship in improving growth under increasing income inequality. Ease of startup and domestic credit from financial sector matter, in the sense that, both factors, under increasing income inequality, are positively associated with growth per capita. We hope that other researchers will explore the robustness of our results in alternative settings and examine the role of other variables that might moderate the pernicious effects of economic inequality on society.

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## Notes

1. In this study, the outcome variable is GDP per capita growth percentage, and is defined by the Organisation for Economic Co-operation and Development (OECD) as

   > one of the core indicators of economic performance. GDP per capita growth can be broken down into growth in labour productivity, measured as growth in GDP per hour worked, and changes in the extent of labour utilisation, measured as changes in hours worked per capita.

   The proposed outcome measure is directly related to the diffusion of earning opportunities across a population and helps assess the extent to which income inequality is lowered through inclusion of, and participation by, workers in economic growth. We refer to this outcome as growth per capita.
2. Startup is defined as a firm during early stages of its operations. Nascent firms are in the prefounding stages that are in the process of collecting resources to start a firm, but do not have sales. After the nascent stage, in the startup stage, firms have low revenues and high costs. Ease of startup refers to the ease in transitioning from nascent to startup phase.
3. Source: https://data.worldbank.org/indicator/IC.REG.COST.PC.ZS
4. http://www.cepii.fr/cepii/en/bdd_modele/bdd.asp
5. Appendix A retrieved on July 31, 2017, from http://www.sciencedirect.com/science/article/pii/S0304387806001830#aep-section-id17
6. Source: http://data.worldbank.org/indicator/SI.POV.GINI
7. In our sample, this measure is correlated at .99 with procedures for males and females or with procedures for males and tends to be somewhat more comprehensive as females face more hurdles in startups.
8. Source: http://data.worldbank.org/indicator/FS.AST.PRVT.GD.ZS
9. Source: http://data.worldbank.org/indicator/IC.BUS.NDNS.ZS
10. Using the margins routine in Stata: "margins, dydx(<moderator>) at, predictor = (<min to max>),"

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